

Conjugate heat transfer at an immersed interface: the scheme as implemented

As-built note — what the solver does today

Abstract

This note describes the conjugate-heat-transfer scheme *as it exists in the code*. It is deliberately narrower than the design note `conjugate_ibm.tex`, which surveys three source methods, derives the exact cut-face flux including its tangential term, works out a corner model, and weighs alternatives. None of that survey is repeated here. What is described here is what runs: one face coefficient, a fluid-fraction-weighted capacity, a time-step bound, a configuration surface with five hard guards, and two diagnostics. Where a term exists in the code but is switched off by default, it is stated as such together with the measurements that decided it (§8); where a piece of the design was never built, it is listed so that the reader is not misled (§9).

1 What is solved

The solid stops being a boundary condition and becomes a real unknown. Fluid and solid carry the same scalar variable, advanced by the same kernel, and differ only through two ratios that are 1 in the fluid by definition:

$$\kappa = \frac{k}{k_f}, \quad C = \frac{(\rho c)}{(\rho c)_f}.$$

In the solver's non-dimensional form, with $D_f = 1/(Re Pr)$ the fluid diffusivity, the transported scalar s obeys

$$C \partial_t s = -\nabla \cdot (\mathbf{u} s) + \nabla \cdot (\kappa D_f \nabla s) + C q, \quad (1)$$

with $\mathbf{u} \equiv \mathbf{0}$ and $\kappa = \kappa_s$, $C = C_s$ inside the body. At the interface the physical conditions are continuity of temperature and of normal flux,

$$s_f = s_s, \quad k_f \partial_n s_f = k_s \partial_n s_s, \quad (2)$$

optionally relaxed to a contact resistance R_c (§3). **Neither condition is imposed anywhere in the code.** They are consequences of a single face coefficient, which is the whole scheme.

2 Geometry: one stored field

The scheme reads exactly one geometric field, and it is one the solver already produced for the turbulence models:

$$\varphi = \pm \text{dwall}, \quad (3)$$

the distance to the surface, *signed* negative inside the solid. It is ghost-inclusive and pointwise exact from the geometry — no halo exchange is needed for it, because every value is computed from the surface, not from neighbours.

The sign comes from the cell-centred immersed-boundary marker: a cell centre lies in the solid exactly when $|\text{coef}(\text{VAR_P})| > 10^{20}$, because `set_ibm_coeff` writes `SOLID/Re` there and grades only fluid-centred cells. Both producers of `dwall` (the file-based path via the case file's `dwall_blocks`, and the analytic path) already supply what is needed; the conjugate mode adds **no new dataset and no case-file format change**.

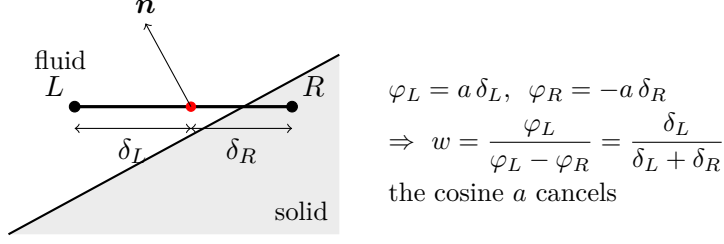


Figure 1: The obliquity lemma. φ stores the perpendicular distance, so the direction cosine a appears in both signed distances and cancels in their ratio: w is the arm fraction at any orientation, with no normal computed.

The obliquity lemma. This is why no normal vector is ever computed. Let the interface be locally a plane with unit normal $\mathbf{n}$, and consider a grid arm along $\mathbf{e}_d$ joining two cell centres at distance h_d , with $a = \mathbf{n} \cdot \mathbf{e}_d$ the direction cosine. Because φ stores the *perpendicular* distance,

$$\varphi_L = a \delta_L, \quad \varphi_R = -a \delta_R,$$

where δ_L, δ_R are the distances *along the arm* from each centre to the interface. The cosine therefore cancels in the ratio:

$$w = \frac{\varphi_L}{\varphi_L - \varphi_R} = \frac{\delta_L}{\delta_L + \delta_R} \quad (4)$$

is the true arm fraction *at any orientation*. That is the entire reason the scheme needs no new data.

3 The face coefficient — the whole scheme

At a face whose two cell centres straddle the interface ($\varphi_L \varphi_R < 0$), the face diffusivity becomes the distance-weighted harmonic mean on w :

$$k_{\text{face}} = \frac{D_f}{\frac{w}{\kappa_L} + \frac{1-w}{\kappa_R} + R_c \frac{D_f}{h_d}} \quad (5)$$

Every other face keeps today's kernel line verbatim, $k_{\text{face}} = \kappa_L D_f$. That is the complete interface treatment: one sign test, one weight, one reciprocal, evaluated inside the flux line the kernel already had.

What it inherits:

- **Exact in one dimension.** It is the series resistance, so it is exact for a grid-aligned interface at *any* cut position, and exact at any orientation wherever the local gradient is interface-normal.
- **Conservative.** It is a face flux — the neighbour subtracts the same number — so $\sum C s \Delta V$ is conserved to round-off.
- **Correct in both limits.** $\kappa_R \rightarrow \infty$ recovers the isothermal wall, $\kappa_R \rightarrow 0$ zero flux.
- **Inert when unused.** An uncut face takes the else-branch, which is the pre-existing arithmetic operand for operand, so a non-conjugate run is bit-identical.

Grazing guard. w is unreliable for arms nearly *tangent* to the surface, where $\varphi_L - \varphi_R \rightarrow 0$. The implementation therefore tests the direction cosine, which (4) supplies for free as $a = |\varphi_L - \varphi_R|/h_d$, and falls back to $w = \frac{1}{2}$ below a threshold. Such arms carry little of the interface flux, so the fallback costs nothing measurable.

Contact resistance. R_c enters (5) as a third resistance in series, which is what it physically is. It is exact: a gate with $R_c = 0.25$ reproduces the analytic two-material solution to 0.0.

4 The discrete scheme

4.1 Face loop

The six face diffusivities of a pressure cell are built with (5) and the flux divergence is formed as usual. Nothing else in the diffusion stencil changes.

4.2 Convection

Convection is hard-masked on solid *and* cut faces, and — this is the part that is easy to get wrong — the skew-symmetric term’s divergence is built from the *same* masked velocities. That is what keeps a uniform scalar exactly preserved, in the solid as well as the fluid.

4.3 Capacity at cut cells

The flux divergence is divided by the cell’s own volumetric capacity. A cut cell physically contains both materials, so the capacity is **fluid-fraction weighted**:

$$C_{\text{cell}} = f + (1 - f) C_s, \quad \text{rhs} = \frac{-\text{conv} + \text{diff} + f q_f + (1 - f) C_s q_s}{C_{\text{cell}}}, \quad (6)$$

with f the fluid volume fraction of the cell. The volumetric source is weighted with the same fraction, as the power density it is.

f is pure geometry, built once at init beside φ from the closed plane-in-box form and stored as `vfrac`; it rides the snapshot so a diagnostic can state the invariant the solver actually conserves. The common approximation $\text{clip}(\frac{1}{2} + \varphi_c/h)$ is not an approximation of that form but its degenerate limit — the one a grid-aligned wall lands in.

Why the weighting is not cosmetic. It cannot change any steady state, because it divides a residual that vanishes there. It is what makes the *transient* second order: with a pointwise capacity a cut cell carries one material’s entire heat capacity, an $O(1)$ error on an $O(h)$ band, i.e. first order. Measured on a two-material eigenmode, against the pre-C3 binary on the same inputs:

	field-error order	decay-rate order
fluid-fraction weighted (6)	1.99, 2.00	1.99, 2.00
pointwise capacity	1.71, 1.09	1.28, 0.54

4.4 What is deliberately absent

There is **no penalization** in the conjugate mode. Its solid cells are unknowns, not boundary values, so the volume-penalization term that the `dirichlet` mode applies is simply not entered. Likewise ν_t is admitted neither in the solid nor at a cut face.

5 Time step

Two findings, both measured, are built into the explicit bound.

The limit is not the maximum material diffusivity. A cut face’s k_{face} can approach $\max(\kappa_L, \kappa_R)$ while feeding the cell on the *other* side, whose capacity is the other material’s. A fluid cell against a $\kappa_s = 1000$ solid therefore sees $1000\times$ the fluid rate *even when* $\alpha_s = \alpha_f$. The rate is consequently assembled from the *actual face coefficients* rather than from $\max(\kappa/C)$ — which is also some $500\times$ less conservative at $w = \frac{1}{2}$.

A cut cell attains the Gershgorin bound. The uniform interior never excites $\rho \leq 2A_{ii}/C_i$; a cut cell does. Its rate is therefore raised: the implementation uses

$$\text{share} = \begin{cases} 2 & \text{at a cut cell,} \\ 6 & \text{otherwise,} \end{cases}$$

i.e. $\text{dt} \times \frac{2}{3}$ *at cut cells only*. The history is worth keeping: the first version used 6 everywhere, C1 halved it to 3, and C2 then measured the default sitting at 96 % of the bound and found an oblique high-contrast case that went to NaN there. At $\text{share} = 2$ the margin is $1.56\times$ and that case runs.

6 Configuration surface

key	meaning
[scalar.N] <code>ibm_wall = conjugate</code>	selects the mode (beside <code>dirichlet</code> , <code>adiabatic</code>)
<code>solid_k</code>	κ_s
<code>solid_rho</code>	C_s
<code>solid_init</code>	initial solid value
<code>solid_source</code>	volumetric source inside the solid
<code>contact_resistance</code>	R_c in (5)
<code>tangential_correction</code>	the C2 term, default off (§8)
[scalar] <code>indicator_interval</code>	the free error indicator
[scalar] <code>heat_interval</code>	the interface-heat diagnostic

`dirichlet` and `adiabatic` were left untouched: re-expressing them through the new arithmetic could not have been bit-exact, and bit-exactness of the existing modes was worth more than the shared code path.

Five hard configuration errors. Each of these silently produces a wrong answer if allowed, so each stops the run with a message naming the fix: `remove_solid = true` (the solid must exist); `refine_body` without `keep_buried` (same); `ibm_value` on a conjugate scalar (there is no imposed value); a `solid_*` key on a non-conjugate scalar; no immersed body at all; and wall functions. A sixth check is geometric rather than configurational: the 2:1 precondition — a cut face may not sit on a coarse/fine block face, because the coefficient is a same-level arm — is *verified* at init, not assumed.

7 Diagnostics

Interface heat (the Nusselt number). The conjugate branch sums the scheme’s *own* face flux over the cut faces, with the penalization column explicitly zero — there is none in this

mode, and `coef_p` is finite in a graded fluid cell, so writing the zero explicitly forestalls a double count. Gated three ways: against a closed form (4.6×10^{-13}), against an independent Python transcription (3.6×10^{-16}), and against a control-volume budget with the flow on that falls like dt^2 and does not move with `niter`.

The error indicator. `indicator_interval` reduces $e_{\text{face}} = h_d s_t / (T_R - T_L)$ over the cut faces (max, rms, count). This is the closed form of the relative error the scheme makes by dropping the tangential term, so it is a genuine *a-posteriori* measurement rather than an assumption, and it is free. It ships enabled.

8 Present but disabled: the tangential correction

The exact cut-face flux carries a second term, $s_t (k_{\text{loc}} - k_{\text{face}})$, which the scheme above drops. It *is* implemented, as its own flux divergence added after the baseline term — so it is inert by construction when off, not by a `+0.0` argument — and it is **default off**. Three independent measurements decided that:

1. a crossover ratio $r^* \approx 0.027$, which the DNS-like $|\nabla_t T|/|\partial_n T| \sim 10^{-2}$ sits below;
2. an $r = \infty$ boundary-value problem in which the corrected *face flux* is exact to 10^{-11} while the corrected *field* is $2.6\times$ worse — a finite-difference divergence wants the face *average*, not the midpoint value;
3. decisively, a cylinder at $\kappa_s = 10^3$ where the correction is $40\text{--}65\times$ worse than dropping the term, because $|\nabla_t T| \propto 2/(1 + \kappa_s) \rightarrow 0$ in the isothermal limit while the discrete estimate of it does not.

The measured fix for (3) — k_{area} , the face area-weighted mean, which removes the paradox exactly on a plane — is recorded but deliberately not implemented: it cannot rescue the feature, because the binding constraint is the s_t estimate on curved interfaces. **Anyone reopening this starts from s_t , not from the multiplier.**

9 Not implemented

For the avoidance of doubt, the following appear in the design note and do *not* exist in the code: the corner/wedge (COCO) model for conducting sharp corners; the escalation back to the full exact flux as a supported mode; and any pluggable local-solution interface. The corner model rests on the same pointwise-flux premise that measurement (2) above falsified, so it should not be started without first settling the s_t estimate.

10 Validation status

Manufactured solutions and identities. A 1D two-material slab with the cut swept through a full cell and κ_s over five decades: $\max |\theta - \theta_{\text{exact}}| \leq 9.7 \times 10^{-15}$ over 28 pairs, and w itself 2.2×10^{-14} against the analytic cut, rebuilt from the case file without the solver. Capacity-independent steady state; contact resistance exact; $\sum C\theta \Delta V$ drift 1.2×10^{-16} relative with the flow on. 1 rank = 4 ranks and CPU = GPU at `max_abs 0` on both geometry paths. Every `ibm_wall` $\neq$ `conjugate` run and every `[scalar] count = 0` run bit-exact against the pre-feature binaries.

Turbulence. A conjugate channel at $Re_\tau = 149$, $Pr = 0.71$, matched to Flageul et al. (2015) in thermal problem, Reynolds number, grid resolution, solid thickness and outer-wall boundary condition, compared against *their published raw data*. Agreement, as a percentage of each quantity’s own peak:

U^+	0.5 %	$\overline{u'v'}$	0.6 %
θ^+	1.1 %	$\overline{u'T'}$	1.1 %
$\overline{u'^2}$	1.0 %	$\overline{v'^2}$	2.2 %
$\overline{w'^2}$	4.3 %	$\overline{T'^2}$	2.2 %

The velocity comparison matters independently: it is the first against a trustworthy reference, so the thermal agreement is not resting on a compensating error in the flow.